

Special Problem Defense

MitoChime

A Machine Learning Pipeline for Detecting PCR-Induced
Chimeric Reads in Mitochondrial Illumina Data

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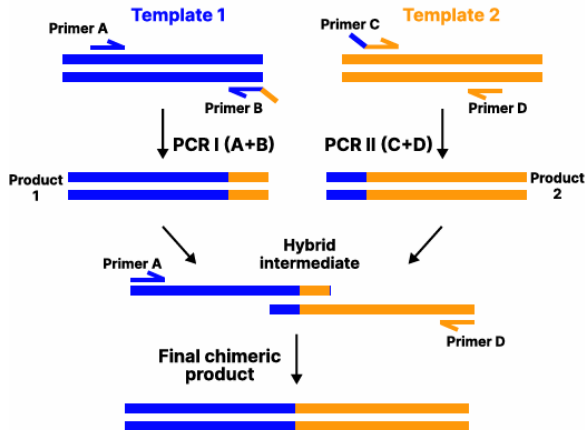
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Four simulated *Sardinella lemuru* mtDNA short reads

Read 1	TCAGCCTTTACCAAGGCCCCACCCTTGCGGCCCTCGCCTGCTTAATTTTACTTTCTTA...
Read 2	TAAGTAAGAACTAGTAGGATCGAGGCCTTCCCGTCTAGATAGGCGTTGGCTTAATTGCA...
Read 3	ATAGGGCAGAAGGCCTAGCATGTTAATGGTAAATAGGAAAACCATTAGAGACGTAAATG...
Read 4	GAGGTGGCGAAAGTCAGCGTGGTTCCCAGTCCAAGGCTCATCAGCAAGACGGTTAGGCT...

Question: Out of these short reads, which ones are chimeric?

The Problem



Schematic illustration of PCR-induced chimera formation through incomplete extension and template switching.

Implications

Assembly quality

- ▶ Fragmented or failed mitochondrial assemblies
- ▶ Reduced reliability of mitochondrial genomics workflows

Research impact

- ▶ Limited downstream analyses in fisheries, population genetics, and evolution
- ▶ Wasted sequencing effort, computing time, and researcher labor

Existing approaches

Tool / Approach	Main focus	Gap for this study
UCHIME / UCHIME2	Amplicon chimeras	Reference/abundance dependent
DADA2	ASV-level filtering	Not read-level mtDNA filtering
CATCh	16S rRNA ensemble ML	Not for mtDNA; inconsistent
ChimPipe	RNA-seq fusion/chimera detection	Not PCR-induced mtDNA artifacts
DL-based tools	DNA sequence motif prediction	Few to no tools for chimera detection
MitoChime	Read-level mtDNA chimeras	Pre-assembly ML/DL filtering

Most existing tools target amplicon, ASV, or RNA-seq chimeras; MitoChime targets PCR-induced mitochondrial short-read artifacts.

Can machine learning detect PCR-induced chimeric reads
before mitochondrial genome assembly?

Input

Simulated clean and chimeric paired-end
Illumina reads from *S. lemuru*

Output

A modular pipeline that predicts and filters
suspected chimeric reads

Scope and limitations

Included

- ▶ *Sardinella lemuru*
- ▶ mitochondrial Illumina reads
- ▶ PCR-induced chimeras
- ▶ simulated labelled datasets

Excluded

- ▶ long-read sequencing
- ▶ NUMTs
- ▶ naturally occurring chimeras
- ▶ broad multi-species validation

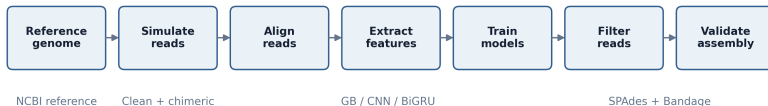
The study prioritized a focused and controlled evaluation setting.

Methodology

From simulated reads to model-based filtering

MitoChime workflow

MitoChime workflow



Overview

The workflow generated labelled reads, extracted read-level evidence, trained models, filtered suspected chimeras, and evaluated downstream assembly behavior.

Data construction

Clean reads

Simulated from the *S. lemur* mitochondrial reference genome.

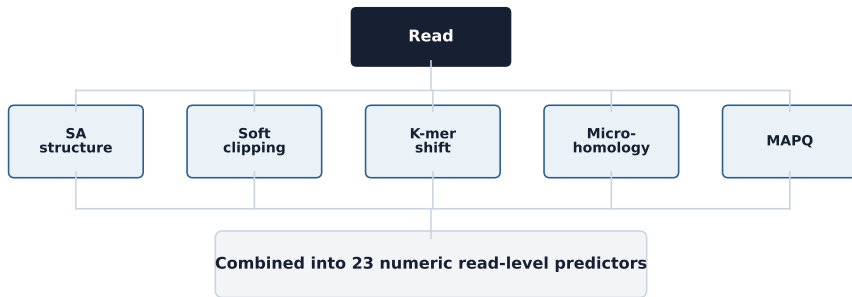
Chimeric reads

Generated by joining non-adjacent mitochondrial segments to mimic PCR template switching.

Labels: **clean** = 0 and **chimeric** = 1

Feature extraction

Read-level feature groups



Feature representation

Each read was converted into a set of 23 numeric predictors describing alignment structure, clipping behavior, sequence composition, microhomology, and mapping quality.

Dataset representation

The labelled reads were prepared in two dataset formats.

read_id	label	seq
NC_039553.1_10001_10492_5:0:0_8:0:0_b77/1	0	GTAAGGCATAGTAAGGCAATGGTGAGGCTGTGTGTGATTGCA
NC_039553.1_10001_10492_5:0:0_8:0:0_b77/2	0	GAGTGAGCCGAATAGGGAATTAGTCCAACCTAAGACTCCTGA
NC_039553.1_10003_10548_5:0:0_8:0:0_f4d/1	0	GTGAGCCGAAAAGGTAATTAGTCCAACCTACGACTCCTGATT
NC_039553.1_10003_10548_5:0:0_8:0:0_f4d/2	0	TGGCTATATAGTGTTGGTGAGGGCTCAGGCCGTTTCCGTTTC
NC_039553.1_10005_10512_6:0:0_6:0:0_c35/1	0	CCGTTGTCCAGTTTAGCCAGGTAAGGCTTAGTAAGGCAATGC
NC_039553.1_10005_10512_6:0:0_6:0:0_c35/2	0	GAGCCGCATAGGGCAATAGTCCAACCTAAGACTCGTGATTTC
NC_039553.1_10008_10523_3:0:0_10:0:0_914/1	0	TCAGCCCCTTTCCGTTGACCAGTTTAGCCAGGTAAGGCTTAC
NC_039553.1_10008_10523_3:0:0_10:0:0_914/2	0	CCGAATAGGGACATAGTCCAACCTAAGACTCCTGATTTCCTGC
NC_039553.1_10012_10513_12:0:0_5:0:0_1340/1	0	TCCGTTGTGCAGTTTAGCCAGGTAAGGCTTAGTAAGGCAATG
NC_039553.1_10012_10513_12:0:0_5:0:0_1340/2	0	ATAGTGAATTAGTCCAAGTTAAGACTCCTGATTTCGGCTCAGA
NC_039553.1_10012_10591_3:0:0_5:0:0_2571/1	0	AGCCAGCATGTAAGAACCACAGCGGGGAGGATCAGGGATC
NC_039553.1_10012_10591_3:0:0_5:0:0_2571/2	0	ATATGGAATTAGTCCAACCTAAGACTCCTGATTTCGGCTCAGA
NC_039553.1_10015_10459_2:0:0_8:0:0_253e/1	0	GTGATTGCAGCAGTTCAAAGTGACTTTTATAGCGCAATCAG
NC_039553.1_10015_10459_2:0:0_8:0:0_253e/2	0	GGGAATTAGTCCAACCTAAGAGTCTGATTTCCGGCTCAGATG
NC_039553.1_10015_10585_10:0:0_11:0:0_d4c/1	0	GGGAATTAGACGAACATAATCTCTGAATTCGGCTCAGATG
NC_039553.1_10015_10585_10:0:0_11:0:0_d4c/2	0	CATGTAAGAACCCACAGCGGGAGGCTAAGGGATCCCTGGC

read_id	label	read_len	mean_bp	ref_name	ref_strand	mapq	cigar	has_si	sa_count	num_seg	sa_diff	config	sa_min_delta_pos	sa_max_delta_pos
NC_039553.1_3_540_8:0:0_6:0:0_a12	0	100	13.000	NC_039553.1	3	0	60 150M	0	0	1	0		0	0
NC_039553.1_4_430_13:0:0_11:0:0_243d	0	150	13.000	NC_039553.1	4	0	60 150M	0	0	1	0		0	0
NC_039553.1_5_405_8:0:0_11:0:0_1440	0	150	13.000	NC_039553.1	5	0	60 150M	0	0	1	0		0	0
NC_039553.1_6_523_8:0:0_9:0:0_82u	0	150	13.000	NC_039553.1	6	0	60 150M	0	0	1	0		0	0
NC_039553.1_9_574_7:0:0_7:0:0_181b	0	150	13.000	NC_039553.1	9	0	60 150M	0	0	1	0		0	0
NC_039553.1_10_501_9:0:0_8:0:0_256b	0	150	13.000	NC_039553.1	10	0	60 150M	0	0	1	0		0	0
NC_039553.1_11_509_8:0:0_11:0:0_a19	0	150	13.000	NC_039553.1	11	0	60 150M	0	0	1	0		0	0
NC_039553.1_12_427_9:0:0_9:0:0_157	0	150	13.000	NC_039553.1	12	0	60 150M	0	0	1	0		0	0
NC_039553.1_12_489_8:0:0_9:0:0_86f	0	150	13.000	NC_039553.1	12	0	60 150M	0	0	1	0		0	0
NC_039553.1_12_593_13:0:0_13:0:0_2375	0	150	13.000	NC_039553.1	12	0	24 150M	0	0	1	0		0	0
NC_039553.1_14_515_7:0:0_7:0:0_20a4	0	150	13.000	NC_039553.1	14	0	60 150M	0	0	1	0		0	0
NC_039553.1_15_525_9:0:0_9:0:0_1e76	0	150	13.000	NC_039553.1	15	0	60 150M	0	0	1	0		0	0
NC_039553.1_17_520_4:0:0_8:0:0_25c	0	150	13.000	NC_039553.1	17	0	60 148MM	0	0	1	0		0	0
NC_039553.1_18_522_8:0:0_9:0:0_c84	0	150	13.000	NC_039553.1	18	0	60 150M	0	0	1	0		0	0
NC_039553.1_18_593_8:0:0_3:0:0_sca	0	150	13.000	NC_039553.1	18	0	60 150M	0	0	1	0		0	0
NC_039553.1_18_482_8:0:0_7:0:0_1e4b	0	150	13.000	NC_039553.1	18	0	60 150M	0	0	1	0		0	0
NC_039553.1_19_516_8:0:0_8:0:0_c1a	0	150	13.000	NC_039553.1	19	0	60 150M	0	0	1	0		0	0
NC_039553.1_20_525_9:0:0_10:0:0_1b37	0	150	13.000	NC_039553.1	20	0	60 150M	0	0	1	0		0	0
NC_039553.1_21_560_10:0:0_9:0:0_189b	0	150	13.000	NC_039553.1	21	0	60 150M	0	0	1	0		0	0
NC_039553.1_23_443_2:0:0_8:0:0_1009	0	150	13.000	NC_039553.1	23	0	60 150M	0	0	1	0		0	0
NC_039553.1_25_571_7:0:0_8:0:0_1932	0	150	13.000	NC_039553.1	25	0	60 150M	0	0	1	0		0	0
NC_039553.1_28_536_8:0:0_8:0:0_182	0	150	13.000	NC_039553.1	28	0	60 150M	0	0	1	0		0	0
NC_039553.1_32_487_8:0:0_8:0:0_2005	0	150	13.000	NC_039553.1	32	0	60 150M	0	0	1	0		0	0
NC_039553.1_34_520_8:0:0_9:0:0_691	0	150	13.000	NC_039553.1	34	0	60 150M	0	0	1	0		0	0
NC_039553.1_34_452_2:0:0_16:0:0_495	0	150	13.000	NC_039553.1	34	0	60 150M	0	0	1	0		0	0
NC_039553.1_36_574_7:0:0_8:0:0_993c	0	150	13.000	NC_039553.1	36	0	60 150M	0	0	1	0		0	0

Feature matrix

For machine learning models

Sequence dataset

For sequence-based deep learning models

Models evaluated

Gradient Boosting

Feature-based model using engineered alignment and sequence-derived predictors.

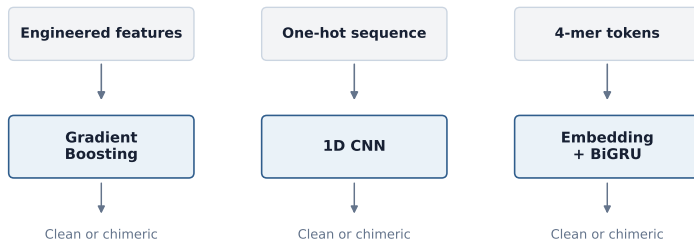
1D CNN

Sequence-based model using one-hot encoded nucleotide reads.

BiGRU

Sequence-based model using ordered k-mer tokens.

Three model representations



Evaluation strategy

Read-level evaluation

- ▶ Accuracy
- ▶ Precision
- ▶ Recall
- ▶ F1-score
- ▶ ROC-AUC

Assembly-level validation

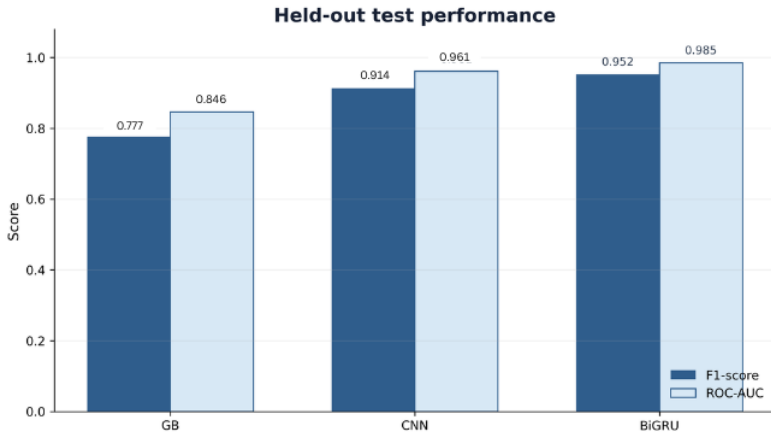
- ▶ Read retention
- ▶ Number of nodes
- ▶ Assembly graph structure
- ▶ Contiguity after filtering

The study evaluated both prediction quality and downstream utility.

Results and Discussion

Classification performance and assembly behavior

Classification performance

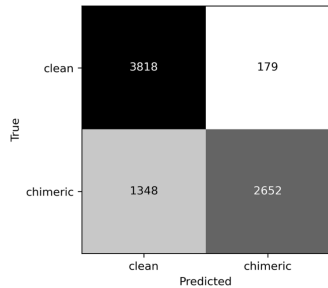


Classification performance

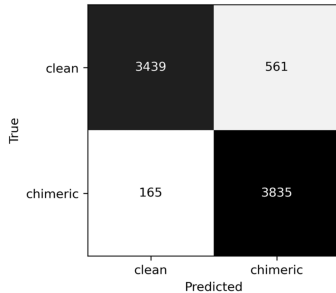
Model	F1-score	ROC-AUC	Runtime
Gradient Boosting	0.7765	0.8459	0.50 min
1D CNN	0.9135	0.9607	1.39 min
BiGRU	0.9523	0.9849	5.51 min

- ▶ **Gradient Boosting** — Conservative filtering; higher read retention
- ▶ **CNN** — Sensitive model; more aggressive filtering
- ▶ **BiGRU** — Highest F1 score; best read-level performance

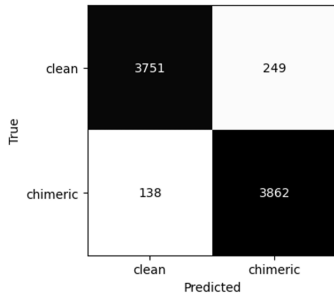
Error patterns



Gradient Boosting

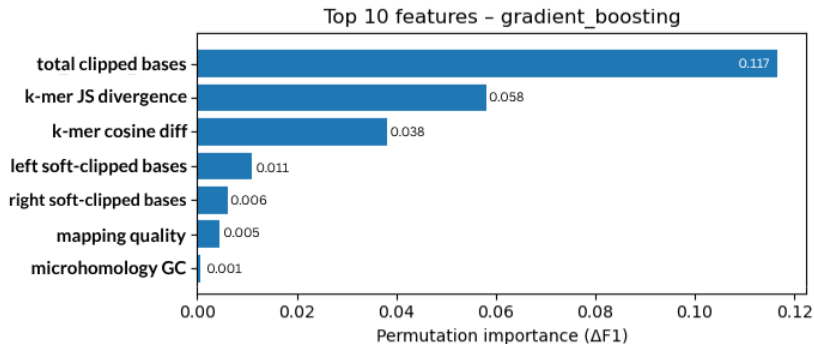


CNN



BiGRU

Feature importance

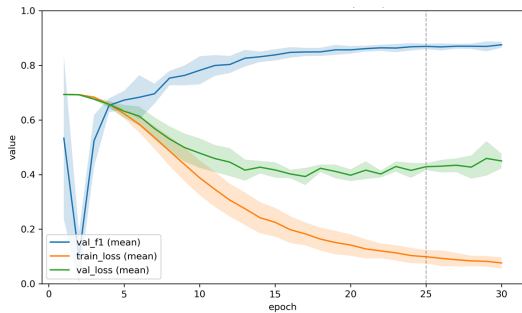


Most informative predictors: **total clipped bases**, **k-mer Jensen–Shannon divergence**, **k-mer cosine difference**, and **left soft clipping**.

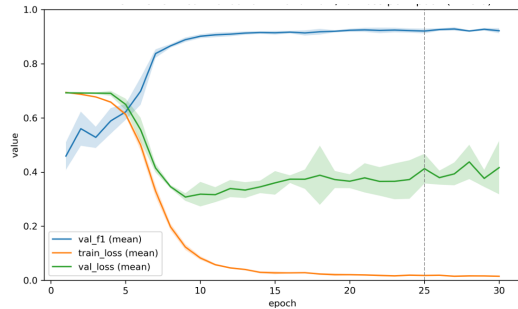
Feature Family Importance

Family	Importance Sum	Importance Mean	nFeatures
Clipping	0.133830	0.033458	4
K-mer jump	0.096264	0.048132	2
Other	0.003676	0.001225	3
Microhomology	0.000347	0.000174	2
SA structure	-0.000859	-0.000066	13

Training Behavior of CNN and BiGRU



CNN



BiGRU

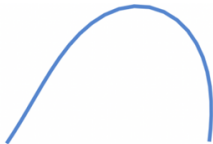
Assembly validation: high coverage



(a) Unfiltered



(b) GB



(c) CNN



(d) BiGRU

20K reads, 50% chimeras

- ▶ Unfiltered assembly showed severe graph complexity.
- ▶ Filtering simplified the graph.
- ▶ CNN produced the simplest graph.
- ▶ GB retained more reads.

Assembly validation: low coverage

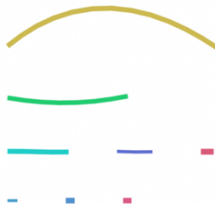


(a) Unfiltered

(b) GB



(c) CNN



(d) BiGRU

3,200 reads, 50% chimeras

At low read depth, aggressive filtering may remove too much read support and cause fragmentation.

Practical implication

The safest model depends on coverage and chimera proportion.

Recommendations: Practical Use

- ▶ Use **Gradient Boosting** when read depth is low.
- ▶ Use **BiGRU** when read-level classification is the priority.
- ▶ **CNN** may be useful in high-coverage and high-chimera conditions.

Recommendations: Future Work

- ▶ Test on real mitochondrial datasets.
- ▶ Evaluate realistic class imbalance.
- ▶ Add varied sequencing-error profiles.
- ▶ Explore hybrid feature-sequence models.

Summary

- ① PCR-induced chimeras can disrupt mitochondrial genome assembly.
- ② Existing tools do not directly address this read-level setting.
- ③ MitoChime detects suspected chimeric reads before assembly.
- ④ BiGRU achieved the strongest classification performance.
- ⑤ Model choice should consider both accuracy and read retention.

MitoChime

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Chimeric Reads in Mitochondrial Illumina Data

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Thank you.